

GREEN INDUSTRIES

Green industries courses offer students opportunities to investigate how to care for and sustainably manage our natural and living resources. Students will explore a variety of areas within green industries, including agriculture, floristry, forestry, horticulture, and/or landscaping.

Students will gain practical skills and knowledge by completing a variety of authentic, industry-relevant activities and projects, such as plant propagation and greenhouse maintenance activities; landscape design and planning exercises; assignments that include farm or forestry management plans; and various construction projects. In addition, students will be introduced to concepts pertaining to biodiversity, environmental sustainability, and natural versus artificial products. Students will also investigate the numerous and varied ways in which green industries and society are interdependent.

Green industries courses will prepare students for working safely in the many fields that are encompassed in this subject area. Students will learn about career opportunities in the green industries and about the Essential Skills and work habits that are important for success in these fields.

The list of approved emphasis areas for green industries can be found at www.edu.gov.on.ca/eng/curriculum/secondary/teched.html.

- Courses in technological education are suitable for use in cooperative education programs and in connection with other forms of experiential learning as well as in programs such as the Specialist High Skills Major (SHSM). For more information, see pages 43–44 of this document.
- For policy guidelines pertaining to multiple-credit courses and emphasis courses, see pages 17–18 of this document.

Green Industries, Grade 11

University/College Preparation

THJ3M

This course enables students to develop knowledge and skills related to agriculture, forestry, horticulture, and landscaping. Students will study the identification, growth, and management of plants and animals and develop process, design, and management skills required in the green industries. Students will also examine social and economic issues related to the green industries, learn about safe and healthy working practices, study industry standards and codes, and will explore postsecondary education programs and career opportunities.

Prerequisite: None

A. GREEN INDUSTRY FUNDAMENTALS

OVERALL EXPECTATIONS

By the end of this course, students will:

- A1.** demonstrate an understanding of species classification and identification and relationships between species and geographical regions;
- A2.** demonstrate an understanding of the effects of biotic and abiotic factors on growth and product quality;
- A3.** develop and evaluate designs or processes for a variety of applications related to the green industries;
- A4.** use mathematical, documentation, research, and communication skills as they apply to the green industries.

SPECIFIC EXPECTATIONS

A1. Species Classification and Geographical Regions

By the end of this course, students will:

- A1.1** distinguish between different plant and/or animal groups on the basis of key identification characteristics, and identify species using both common names and scientific classifications (*e.g., annuals and perennials; native and non-native plants; major types, species, and varieties of trees, shrubs, flowering plants, and crops; animal breeds*);
- A1.2** identify geographical regions on the basis of classification criteria relevant to the green industries (*e.g., forest type, hardiness, agricultural use, ease of cultivation, water features*);
- A1.3** explain the relationships between the characteristics of different geographical regions and the key desirable characteristics of plant and/or animal groups within them (*e.g., relationship of plant and animal characteristics to available heat, moisture, light, shelter, and food*).

A2. Factors Affecting Growth and Product Quality

By the end of this course, students will:

- A2.1** describe how abiotic factors (*e.g., air quality, temperature, nutrients, water, topography, handling procedures*) affect the growth of various plant and/or animal species and the quality of products derived from them;

- A2.2** explain biological processes that are essential to the propagation, development, and health of plants and/or animals and the quality of products derived from them (*e.g., reproduction, respiration, photosynthesis, transpiration, post-harvest physiology*);

- A2.3** identify a variety of pests and diseases (*e.g., bacteria, viruses, moulds, fungi, insects, animals*) and explain their effects on the health of plants and/or animals and the quality of products derived from them.

A3. Designs and Processes

By the end of this course, students will:

- A3.1** demonstrate an understanding of and apply the steps in a design process (see pp. 22–23) to a variety of requirements in the green industries (*e.g., creation of forest management plans, environmental farm plans, urban landscape designs, hydroponic system designs*);
- A3.2** explain fundamental operational processes that are commonly used in the green industries (*e.g., single animal management, crop location and rotation, crop scheduling, event planning, nutrient and waste management, composting, select cutting, timber cruise*);
- A3.3** identify a variety of structures used in the green industries (*e.g., mills, barns, different greenhouse styles, store layouts*), and explain how their structural features relate to their functions;

A3.4 demonstrate an understanding of correct procedures for the care and handling of plants and/or animals (*e.g., propagating, pruning, transporting, watering, feeding, fertilizing, removing bark*).

A4. Technological and Mathematical Literacy and Communication Skills

By the end of this course, students will:

A4.1 demonstrate an understanding of terminology used in the green industries and use it correctly in oral and written communication (*e.g., sustainability, coniferous, massing flower, flagstone, organic*);

A4.2 use effective documentation practices to record and track important information related to green industry operations (*e.g., preparing invoices, recording fertilizer and pesticide use, completing maintenance records, maintaining crop management records, documenting Hazard Analysis and Critical Control Point [HACCP] activities*);

A4.3 describe commercial and technical issues of current significance in the green industries (*e.g., adoption of innovative technologies, marketing boards versus open markets, food safety testing*), using information from industry sources (*e.g., trade publications and websites, marketing boards, growers' associations*);

A4.4 perform metric and imperial unit conversions and other calculations as required for a variety of green industry applications (*e.g., land areas, yields, green log weights, diet analyses, invoices*).

B. GREEN INDUSTRY SKILLS

OVERALL EXPECTATIONS

By the end of this course, students will:

- B1.** demonstrate an understanding of and apply design and production practices that are commonly used in the green industries;
- B2.** apply management strategies for assessing and controlling biotic and abiotic factors that affect plant and/or animal quality;
- B3.** demonstrate competence in technical skills related to specific applications and tasks within the green industries.

SPECIFIC EXPECTATIONS

B1. Design and Production

By the end of this course, students will:

- B1.1** implement a production process or procedures according to a design or plan (*e.g., timber cruise, stand inventory, landscape construction, crop rotation, mixed animal farming, selective breeding*);
- B1.2** utilize a management plan for a specific application related to the green industries (*e.g., forest management plan, nutrient management plan, site layout plan, crop rotation plan, annual work plan, business plan, five-year operational plan*);
- B1.3** demonstrate an understanding of and apply techniques related to the propagation and maintenance of a variety of plant and/or animal species and the post-harvest handling of plant products (*e.g., techniques related to crop, mammal, and poultry production, sexual and asexual plant production, shrub rejuvenation, rose processing, care of selected local tree species*);
- B1.4** demonstrate an understanding of and apply techniques or processes that promote biodiversity, increase ecosystem function, and reduce maintenance requirements (*e.g., planting native species, mulching, establishing natural habitat*);
- B1.5** demonstrate an understanding of and apply marketing techniques that are commonly used in the green industries (*e.g., product displays, flyer and poster advertising, Internet sales*).

B2. Plant and Animal Management Strategies

By the end of this course, students will:

- B2.1** apply a variety of methods to monitor and assess biotic factors that affect plant and/or animal quality (*e.g., weed identification, regular animal health inspections, plant quality inspections, pest scouting, post-harvest tracking of product freshness and quality*);
- B2.2** apply a variety of methods to monitor and assess abiotic factors that affect plant and/or animal quality (*e.g., nutrient balance analysis, soil testing, plant tissue analysis, monitoring growing degree days [GDDs], form defect analysis, water testing*);
- B2.3** apply a variety of techniques to control pests and reduce plant and/or animal defects (*e.g., maintenance or enhancement of natural barriers to control pest migration, animal quarantine, log hydration, integrated pest management*).

B3. Technical Skills

By the end of this course, students will:

- B3.1** demonstrate competence in the technical skills required to complete a variety of tasks in the green industries (*e.g., operating power tools and machinery, rigging loads, hardscaping, pesticide application*);

B3.2 construct a variety of structures or products that are used or produced in the green industries (*e.g., trellises, scaffolds, containers, raised beds, fences*);

B3.3 demonstrate competence in related technical skills (*e.g., using GPS equipment, welding, wiring and making electrical repairs, operating and maintaining small engines, making orthographic drawings, using computer applications*) that are required to complete a variety of green industry projects.

C. TECHNOLOGY, THE ENVIRONMENT, AND SOCIETY

OVERALL EXPECTATIONS

By the end of this course, students will:

- C1.** analyse the impact of the green industries on the environment and describe ways of minimizing harmful effects;
- C2.** analyse social and economic relationships and issues involving the green industries.

SPECIFIC EXPECTATIONS

C1. Technology and the Environment

By the end of this course, students will:

- C1.1** demonstrate an understanding of ecological relationships and processes (*e.g., food webs, symbiotic relationships, ecological succession, nutrient flows and cycles, habitat and species diversity*) that can affect or be affected by green industry operations;
- C1.2** analyse the effects of green industry activities on the environment in the past and in the present (*e.g., logging practices, irrigation, fertilization, pest control, nutrient and waste management*);
- C1.3** assess the advantages and disadvantages of using natural rather than manufactured materials or products in green industry activities (*e.g., natural fertilizers and pest control methods rather than chemical fertilizers and pesticides, real flowers rather than artificial flowers, real grass rather than artificial turf, untreated rather than pressure-treated lumber*);
- C1.4** identify sustainable practices and guidelines that are currently applied within the green industries or may be applied in the future (*e.g., environmental farm planning, integrated pest management, xeriscaping, forest regeneration, low-till cultivation*);
- C1.5** explain the environmental implications (*e.g., effects on landfill lifespan and water and air quality*) of using particular materials, products, processes, and disposal methods (*e.g., recycling, reusing, composting, growing genetically modified crops, organic farming, various disposal methods for invasive plants*).

C2. Technology and Society

By the end of this course, students will:

- C2.1** describe linkages between local communities and economies and the green industries (*e.g., consumer needs and green industry goods and services, green industries as sources of jobs and tax revenues, communities as providers of services to green industries and as regulators of green industry activities*);
- C2.2** analyse societal issues relating to the green industries, and identify ways of resolving them, taking a variety of perspectives into account (*e.g., effects on Aboriginal hunting and harvesting territories, land use conflicts such as parkland versus commercial development, property rights and municipal landscape management, animal welfare, rights of migrant workers, fair trade concerns relating to imported agricultural or floral products, fuel ethanol versus food production*).

D. PROFESSIONAL PRACTICE AND CAREER OPPORTUNITIES

OVERALL EXPECTATIONS

By the end of this course, students will:

- D1.** demonstrate an understanding of and apply safe working practices as they relate to the green industries;
- D2.** demonstrate an understanding of the business and regulatory environment of the green industries;
- D3.** identify careers in the green industries, and describe the skills, education, and training required for entry into these occupations.

SPECIFIC EXPECTATIONS

D1. Health and Safety

By the end of this course, students will:

- D1.1** identify the personal protective clothing and equipment needed to perform various green industry tasks safely, and use as required to ensure their own and others' safety in the work environment (*e.g., eye and ear protection; hand, head, and foot protection; sun protection; equipment guards*);
- D1.2** demonstrate an understanding of environmental and site-related hazards (*e.g., land conditions; weather conditions; crew competence and organization; presence of utility lines, glass structures, hanging limbs, chicots*) and apply appropriate safety measures for avoiding them (*e.g., roping off an area, setting up caution signs, removing hazards, implementing traffic control measures*);
- D1.3** demonstrate an understanding of and apply safe procedures for using and maintaining materials, tools, and equipment (*e.g., avoid moving parts and pinch points; perform a circle check of vehicles and equipment; check condition of materials, hydraulic systems, and protective equipment; check oil and fuel levels*);
- D1.4** demonstrate an understanding of and apply safe procedures for handling plants and/or animals (*e.g., dethorning plants, using ergonomic lifting techniques or devices, using hand protection, securing loads correctly for transport, understanding animal perception, avoiding actions that startle animals, using chutes and restraining devices*);
- D1.5** identify potentially hazardous situations in the workplace by conducting and documenting personal and workplace safety audits;

D1.6 identify sources of information about workplace hazards and how to avoid them (*e.g., Workplace Hazardous Materials Information System [WHMIS], Passport to Safety*);

D1.7 outline and comply with legislation for protecting the health and safety of workers in the green industries (*e.g., Occupational Health and Safety Act, local by-laws, fire prevention regulations*).

D2. Business and Regulatory Environment

By the end of this course, students will:

- D2.1** identify a variety of industry-related organizations (*e.g., local growers' associations; government departments; non-governmental organizations*), and demonstrate an understanding of their role with respect to the green industries;
- D2.2** explain principles, concepts, and practices related to the marketing and distribution of commodity products and/or services (*e.g., product branding; advertising approaches such as health- and lifestyle-related campaigns; market opportunities and challenges such as exports and international/interprovincial competition; distribution considerations such as shipping of live plants and animals, warehouse requirements, and product life*);
- D2.3** outline and comply with legislation and guidelines governing the quality and safety of green industry products and services (*e.g., product quality regulations, grading standards, inspection requirements, voluntary guidelines such as the Landscape Ontario guidelines for the landscaping industry*).

D3. Career Opportunities

By the end of this course, students will:

- D3.1** describe careers in the green industries (*e.g., landscape architect, forest manager, horticulturalist, farm manager, turf manager, botanist, veterinarian*) and the education, training, and certification required for entry into these occupations;
- D3.2** identify ways of acquiring knowledge and experience in green industry occupations (*e.g., through part-time work experience, cooperative education, guest speakers, field trips, job shadowing*);
- D3.3** identify the size and composition of the labour force and the career opportunities available in the green industries in their local and regional communities;
- D3.4** identify groups and programs that are available to support students who are interested in pursuing non-traditional career choices in the green industries (*e.g., mentoring programs, virtual networking/support groups, specialized postsecondary programs, relevant trade/industry associations*);
- D3.5** demonstrate an understanding of and apply the Essential Skills that are important for success in the green industries, as identified in the Ontario Skills Passport (*e.g., reading text, writing, document use, computer use, oral communication, numeracy, thinking skills*);
- D3.6** demonstrate an understanding of and apply the work habits that are important for success in the green industries (*e.g., working safely, teamwork, reliability, initiative, customer service*);
- D3.7** maintain an up-to-date portfolio that includes pieces of work and other materials that provide evidence of their skills and achievements in areas related to the green industries (*e.g., Passport to Safety certificate, technical reports, competency checklists, Ontario Skills Passport Work Plan*), and explain why having a current portfolio is important for career development and advancement.